Vishnu Vardhan

Full Stack AI Engineer

9182177915 | repanavishnuvardhan67664@gmail.com | LinkedIn | GitHub | Portfolio | Bengaluru, Karnataka, India

Professional Summary

Full Stack AI Engineer who ships production LLM and agentic systems, not just notebooks — multi-agent orchestration, RAG pipelines, and the full-stack infrastructure (React, FastAPI, MySQL) to run them at scale. Nearly **1 year of internship experience** taking AI systems from prototype to production: cutting latency by up to **30–40%**, scaling API throughput **3**×, and hardening pipelines to zero critical CVEs. Comfortable owning the full stack end-to-end — model integration, vector search, secure backend services, and deployment.

Experience

The Developers Arena — Hyderabad, Telangana, India
Python Developer Intern — Nov 2025 – May 2026

- Architected **6+ production-grade REST APIs** with **Flask and FastAPI** using **modular MVC design patterns**, enabling seamless third-party integrations and reducing average response latency by **30%** across live client deployments.
- Built **end-to-end AI backend pipelines** integrating **OpenAI/Gemini LLMs, LangChain agents, and RAG** with **Pinecone/ChromaDB vector stores** — delivering intelligent **document Q&A, semantic search, and automated reasoning** features shipped to production.
- Engineered **optimized MySQL schemas** with normalized table design, composite indexing, and stored procedures — cutting query execution time by **40%** and sustaining reliable performance under high-volume concurrent read/write workloads.
- Scaled **API throughput by 3**× by integrating **Apache Kafka event streaming** for real-time data ingestion and implementing **Redis caching, Celery async task queues, and connection pooling** — sustaining **sub-200ms p95 response times** under **500+ concurrent request** load tests.
- Secured backend infrastructure against **OWASP Top 10** attack vectors by integrating **JWT/OAuth2 auth flows, RBAC, AES-256 encryption, and parameterized queries** — achieving **zero critical CVEs** across all API-facing services and blockchain endpoints.
- Automated **ML data preprocessing pipelines** using **Pandas and NumPy** across **50K+ record datasets** — standardizing feature engineering workflows and reducing model training preparation time by **35%**.
- Refactored monolithic backend into **Dockerized microservices** with **Kafka-driven inter-service messaging, Web3.py blockchain audit trails, and RESTful contracts** — reducing deployment downtime by **50%** and enabling **zero-downtime rolling updates** across high-traffic components.
- Sustained **95%+ sprint velocity** across **8+ Agile/Scrum cycles** using **Git flow, GitHub PRs, and Jira** — coordinating across backend, AI/ML, blockchain, QA, and DevOps squads. **View Certificate**

Infotact Solutions — Bengaluru, Karnataka, India
Software Engineer Intern — Sep 2025 – Dec 2025

- Designed and delivered **2+ production-ready full-stack MERN applications** (**React.js, Node.js, Express.js, MongoDB**) featuring **responsive UI/UX** and **RESTful backend architecture** supporting real-world client workflows.
- Engineered **secure, scalable REST APIs** with **JWT-based authentication, Role-Based Access Control (RBAC), and server-side input validation** — ensuring **zero unauthorized access** across all frontend-backend communication channels.
- Optimized **application performance by 35%** through **MongoDB indexing strategies, query optimization, aggregation pipeline tuning, and React component refactoring** — reducing page load time and improving user experience.
- Implemented **CI/CD-ready development workflows** using **Git Flow branching, GitHub Pull Requests, and structured peer code reviews** — enforcing **DRY principles, modular architecture, and SOLID design patterns** across the codebase.
- Collaborated within an **Agile/Scrum team** across **planning, development, testing, and deployment phases** — consistently delivering features on schedule with **high code quality and minimal defect rate**. **View Certificate**

Skills

Programming Languages: Python, JavaScript
Frontend Development: React.js, Next.js, HTML, CSS, Responsive UI Design
Backend Development: Flask, FastAPI, Node.js, Express.js, REST APIs, Authentication, Scalable Backend Services
Databases: MySQL, SQL, NoSQL (MongoDB), Vector Databases (Pinecone, ChromaDB)
Artificial Intelligence / Machine Learning: LangChain, LangGraph, LLM Integration,Prompt Engineering, RAG, Scikit-learn, NLP,Computer Vision
Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD Pipelines, Deployment Automation, Kafka, Redis, Celery
Tools & Platforms: Git, GitHub, Postman, VS Code, Jira, n8n, Jupyter Notebook, Vercel, Netlify
Methodologies: Agile, Scrum, Software Development Life Cycle (SDLC)

Projects

N45 Agentic Builder | *Internship Project — Next.js, React, TypeScript, FastAPI, Python, MySQL, LLM, LangGraph, LangChain, RAG, Vector DB, AW*

- **Business problem:** enterprises lacked a unified visual platform to build, orchestrate, and monitor multi-agent AI workflows — relying instead on fragmented scripts and manual integrations.
- **Architecture:** contributed to a production-ready **Enterprise AI Agent Development Platform** featuring a drag-and-drop **Workflow Builder (React Flow)** and **multi-agent orchestration** for parent–sub-agent collaboration.
- Worked on **workflow validation, versioning, autosave, deployment, and real-time execution monitoring** for reliable end-to-end workflow lifecycle management.
- Helped integrate **multiple LLM providers** (**OpenAI, Gemini, Anthropic, Groq, Ollama, Azure OpenAI**) via configurable AI profiles, with **Knowledge Base (RAG)** support for document ingestion and semantic retrieval.
- Supported implementation of **JWT authentication, RBAC, multi-tenant workspace architecture, and encrypted credential management** for enterprise-grade security.
- Assisted in connecting enterprise tools (**Google Workspace, Slack, GitHub, Jira, Notion, REST/GraphQL APIs, webhooks**) with **analytics dashboards and audit logging** for full observability.
- **Result:** platform enabled enterprises to build and scale AI agent workflows without custom infrastructure — cutting integration overhead across LLM providers and tools.

Agentic HR – AI-Powered Recruitment Pipeline | *Python, LangGraph, FastAPI, Groq LLM, Twilio, Google Calendar API, Gmail API, MongoDB*

- **Business problem:** manual recruitment coordination across shortlisting, screening, scheduling, and onboarding was slow and error-prone for HR teams.
- **Architecture:** designed an **8-node multi-agent pipeline** on **LangGraph StateGraph**, automating the full recruitment lifecycle from resume screening to onboarding with **3 human-in-the-loop approval gates** for reliability and trust.
- Built a **Resume Shortlister Agent** (**Groq LLM, Llama 3.3 70B**) parsing PDF/DOCX resumes and ranking the top **15 candidates** against job descriptions with match scores and rationale.
- Built a **Pre-Screening Call Agent** (**Twilio voice + Edge TTS**) running autonomous live phone screens, extracting **5 structured data points** per candidate (CTC, availability, job-change intent).
- Engineered an **Interview Scheduler Agent** (**Google Calendar + Gmail APIs**) that scans availability, books the first free 1-hour slot, and generates Meet links with zero manual coordination.
- Built an **Onboarding Agent** that auto-sends personalized welcome emails, closing the loop on the full hire-to-onboard workflow.
- **Result:** fully automated an end-to-end HR workflow; scoped production upgrades including low-latency voice streaming and a **Background Verification Agent**. **GitHub Repo**

Online Fee Management System | *HTML, CSS, JavaScript, Python, Flask, MySQL, AI (LangChain, LLMs, RAG, Vector DB)*

- **Business problem:** manual fee collection at **Aditya Degree College** caused delays in fee tracking, reporting, and student query handling.
- **Architecture:** designed and shipped a full-stack AI-powered platform (**Flask REST APIs, MySQL, responsive UI**) to digitize and automate fee operations end-to-end for a live institutional client.
- Delivered **secure admin authentication**, **automated fee tracking**, and **digital receipt generation**.
- Integrated **LLMs + RAG** for intelligent fee queries, summaries, and administrative assistance — reducing turnaround on routine student queries.
- **Result:** cut manual workload and improved fee-management efficiency with a real-time analytics dashboard.
- **Next steps scoped:** multi-agent automation layer (**n8n**) to orchestrate fee reminders, receipt workflows, and admin task handling. **GitHub Repo**

Education

Aditya Degree College — B. Kotha Kota
Bachelor of Computer Applications (BCA); CGPA: 9.05/10 — 2026